

MATH 340

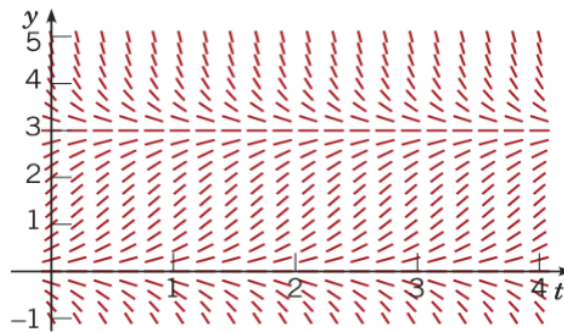
Quiz 1 Solutions

February 7, 2025

1. (5 points) True or False

- (a) ~~(T)~~ (F) The constant function $y'(x) = \pi$ is a solution of the ODE $y'(x) = x \sin^2(y)$.
- (b) ~~(T)~~ (F) For the initial value problem $\begin{cases} x'(t) = 2x + e^t \\ x(0) = 0 \end{cases}$, the solution is $x(t) = e^{2t} - e^t$.
- (c) (T) ~~(F)~~ The ODE $x'(t) + \sin(x(t)) = 0$ is linear.
- (d) (T) ~~(F)~~ The ODE $x'(t) + \sin^2(x(t)) = 0$ is of order 2.
- (e) ~~(T)~~ (F) The ODE $x'(t) + 2tx(t) = 0$ is linear.

2. (5 points) The direction field of an ODE is shown below.



- (a) Circle the differential equation that corresponds to the direction field. Briefly explain why.

$$y' = y(y - 3); \quad \boxed{y' = y(3 - y)}; \quad y' = y - 3; \quad y' = y.$$

We clearly have 2 zeros which are $y = 0$ and $y = 3$ so it has to be one of the first 2 options. Then we see that in order to match the sign of y' we need to pick the second.

- (b) Based on the diagram identify two equilibrium points of the ODE (i.e. where the function stays constant).

The equilibrium solutions are $y = 0$ and $y = 3$.

3. (10 points) Consider the initial value problem $\begin{cases} Q'(t) + \frac{r}{100}Q(t) = \frac{r}{4} \\ Q(0) = Q_0 \end{cases}$ with two positive parameters r, Q_0 (this means they are constant!).

- (a) Find the solution to the IVP.
- (b) Find the limit of the solution as $t \rightarrow \infty$. (That is if $Q_{IVP}(t)$ is the solution you found, then compute $\lim_{t \rightarrow \infty} Q_{IVP}(t)$).
- (a) This is of the form $\frac{dy}{dt} = ay - b$ so we can use the general formula to get:

$$Q_{IVP}(t) = \frac{-\frac{r}{4}}{-\frac{r}{100}} + \left(Q_0 - \frac{-\frac{r}{4}}{-\frac{r}{100}} \right) e^{-\frac{r}{100}t} = 25 + (Q_0 - 25)e^{-rt/100}.$$

- (b) Notice that $(Q_0 - 25)e^{-rt/100} \rightarrow 0$ as $t \rightarrow \infty$, because the exponential goes to 0.

$$\lim_{t \rightarrow \infty} Q_{IVP}(t) = 25.$$